

Literature Review: “Reel or Real? A Context-Aware Deepfake Detection and Verification Framework”

1. Introduction

The generation of synthetic video, image, or audio content using artificial intelligence has shifted from technical novelty to one of the most serious challenges for information integrity. Research since 2020 shows a dramatic rise in both the realism of deepfakes and the difficulty of detecting them (Heidari et al., 2023). As social media and political communication rely increasingly on audiovisual media, the ability to authenticate digital content has become essential. Scholars agree that while deep-learning detection models have advanced quickly, weaknesses remain in generalisation, multimodal understanding, and interpretability (Gong and Li, 2024). This project therefore focuses on building a context-aware detection and verification framework that integrates machine-learning accuracy with transparent, user-centred design.

2. Public Awareness & the Social Dimension

Public concern has outpaced technical understanding. Sippy et al. (2024) surveyed more than 1,400 UK adults and found that although 90 per cent expressed concern about deepfakes, fewer than one in ten could confidently identify them. Similar perception studies across Europe and Asia report high anxiety but low literacy in recognising manipulated media (Ahmed et al., 2022; Kwon et al., 2021). Respondents typically associate deepfakes with celebrity or political scandals but underestimate subtler threats such as identity fraud or misinformation micro-targeting.

These findings demonstrate that detection technology must be trustworthy and explainable to non-experts. Researchers such as Lyu (2020) and Heidari et al. (2023) emphasise that without transparent reasoning, even accurate detectors fail to inspire confidence. The social context therefore justifies the development of software tools that communicate why a video is fake rather than simply labelling it as such.

3. Evolution of Deepfake Detection Techniques

The earliest detection systems relied on convolutional neural networks (CNNs) to identify visual anomalies such as blurring, colour mismatch, or inconsistent blinking (Pan et al., 2020). Comparative studies testing VGG, ResNet and DenseNet models reported accuracies between 93 and 97 per cent on curated datasets (Ahmed et al.,

2022). However, these models were often over-fitted to specific manipulations and degraded sharply when exposed to new forms of forgery.

To overcome this, research diversified toward transformer-based and temporal architectures capable of capturing motion cues across frames (Gong and Li, 2024). Hybrid CNN–transformer approaches now achieve stronger cross-dataset performance, while lightweight networks such as MobileNet have been proposed for mobile deployment (Pan et al., 2020). Nevertheless, the field continues to struggle with generalisation, motivating the design of flexible frameworks rather than single-purpose classifiers.

4. Dataset Diversity & the Realism Challenge

The quality of datasets defines the reliability of detection models. Early benchmarks like FaceForensics++ and the Facebook Deepfake Detection Challenge (DFDC), were groundbreaking but relied on synthetic fakes with obvious artefacts (Lyu, 2020). Newer corpora focus on realism, diversity, and multimodality.

The KoDF dataset introduced 175,000 fake clips of Korean subjects, correcting Western-centric bias (Kwon et al., 2021), while WildDeepfake collected genuine online forgeries exhibiting uncontrolled lighting and compression (Zi et al., 2020). These showed that cross-dataset accuracy could drop by 20 points when training and testing sources differ.

Khalid et al. (2021) further advanced the field with FakeAVCeleb, the first large-scale audio-video dataset, revealing that fusing modalities increased F1-score to 94.3 per cent, well above video-only detection. Other contributions such as KoDF-Mix (2021–22) and WildDF 2023 continue this trend toward inclusive, realistic benchmarks. The consensus is that robust detectors must learn from diverse, socially representative data, a key design principle of this project’s training strategy.

5. Explainability & Trust in AI Detection

Accuracy alone is insufficient. Lyu (2020) warns that black-box neural networks lack the interpretability required for journalism, law, or policy enforcement. Subsequent studies propose attention maps and gradient-based visualisations to highlight tampered regions (Heidari et al., 2023). Transformer attention layers have also been repurposed to expose which facial areas influence model decisions.

Recent interdisciplinary research links explainability to user trust: when users can see why content is flagged, they are more likely to believe the system (Sippy et al., 2024). Ethical reviews emphasise transparency, fairness, and protection of participant data (Gong and Li, 2024). The proposed framework therefore integrates

explainable outputs such as heatmaps, confidence scores, and metadata context to promote informed judgement rather than blind acceptance of AI verdicts.

6. Emerging Approaches & Gaps

The literature reveals four dominant research directions:

- a. Hybrid Architectures: combining CNNs, transformers and temporal modules to exploit both spatial and motion information (Ahmed et al., 2022; Gong and Li, 2024).
- b. Multimodal Detection: integrating audio, visual and even physiological cues such as heart-rate signals (Khalid et al., 2021; Heidari et al., 2023).
- c. Adversarial Robustness: countering attacks that deliberately disguise forgeries (Kwon et al., 2021; Zi et al., 2020).
- d. Ethical Governance: ensuring consent, fairness, and accountability in dataset creation (Sippy et al., 2024).

Despite progress, four gaps persist across all reviewed sources:

- Generalisation: detectors trained on one dataset struggle on unseen manipulations.
- Multimodality: few models fully exploit cross-modal correlations.
- Transparency: interpretability techniques remain under-used in deployed systems.
- Usability: most detectors are laboratory prototypes, not public-facing software.

Addressing these inter-related gaps forms the foundation of this research. By combining algorithmic rigour with user-centred design, the framework seeks to bridge the divide between technical performance and practical trustworthiness.

7. Conclusion & Research Justification

Over the last five years, deepfake detection research has matured from experimental CNN classifiers to sophisticated multimodal, transformer-based systems. Yet the literature shows that even state-of-the-art models falter when facing new forgeries, compressed social-media videos, or public scepticism. Dataset innovation and architectural advances have narrowed the gap but not closed it.

The collective evidence among all academic sources indicates that the next stage of progress requires context-aware, explainable, and ethically grounded frameworks capable of integrating heterogeneous data and presenting intelligible results to end users. This project directly responds to that need, merging the strengths of deep-learning with the principles of software engineering to deliver a system that is technically robust, transparent, and socially responsible, contributing both to scientific understanding and to the restoration of digital trust.

8. References

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